LIM-5.B.1 The limit of an approximating Riemann sum can be interpreted as a definite integral.
LIM-5.B.2 A Riemann sum, which requires a partition of an interval I, is the sum of products, each of which is the value of the function at a point in a subinterval multiplied by the length of that subinterval of the partition.
LIM-5.C.1 The definite integral of a continuous function f over the interval $[a, b]$, denoted by $\int_a^b f(x)dx$ is the limit of Riemann sums as the widths of the subintervals approach 0. That is, $\int_a^b f(x)dx = \lim_{\max \Delta x_i \to 0} \sum_{i=1}^{n} f(x_i^*)\Delta x_i$, where n is the number of subintervals, Δx_i is the width of the ith subinterval, and x_i^* is a value in the ith subinterval.
LIM-5.C.2 A definite integral can be translated into the limit of a related Riemann sum, and the limit of a Riemann sum can be written as a definite integral.
FUN-6.A.1 In some cases, a definite integral can be evaluated by using geometry and the connection between the definite integral and area.

6-3 Riemann Sums, Summation Notation, and Definite Integral Notation

Recall Summation Notation (or Sigma Notation)

The sum of the first n terms of a sequence is represented by the **summation notation**

$$\sum_{k=1}^{n} a_k = a_1 + a_2 + a_3 + a_4 + \cdots + a_n$$

n is the **upper limit of summation**

k is the **index of summation**

1 is the **lower limit of summation**

Get Ready...

Expand and evaluate the following expressions:

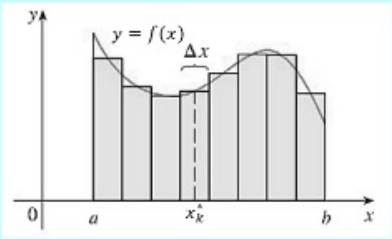
A) $\sum_{k=1}^{3} (k^2 + 1)$ $= 2 + 5 + 10 = 17$

B) $\sum_{k=2}^{4} (3k - 1)$ $= 5 + 8 + 11 = 24$

In sections 6-1 and 6-2 we discussed area under a curve, Riemann sums, and definite integrals. Let's explore this in more detail. Riemann sums will approximate the area under a curve but cannot give you the actual value. The only way to find the actual area under a curve is to use Calculus.

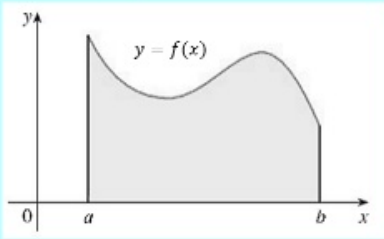
Riemann Sum (area of rectangles)

$$\sum_{k=1}^{n} f(x_k^*)\Delta x$$

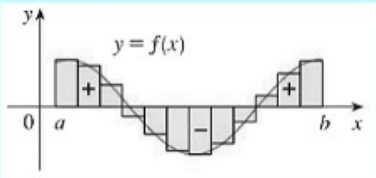


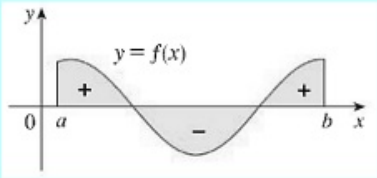
Integral

$$\int_a^b f(x)dx$$



If $f(x)$ takes on both positive and negative values, then the Riemann sum is the sum of the areas of the rectangles that lie above the x-axis and the *negatives* of the areas of the rectangles that lie below the x-axis. When we take the limit of such Riemann sums, we get the definite integral. A definite integral can be interpreted as a **net area**, that is, a difference of areas:





If we take an infinite number of rectangles, a Riemann sum is equivalent to the integral of the function over the interval:

$$\lim_{n \to \infty} \sum_{k=1}^{n} f(x_k^*)\Delta x = \int_a^b f(x)dx$$

The symbol $\int$ was introduced by Leibniz and is called an **integral sign**, $f(x)$ is called the **integrand**, a and b are called the **limits of integration**—a is the **lower limit** and b is the **upper limit**. The symbol dx simply indicates that the independent variable is x. The procedure of calculating an integral is called **integration**.

But as we've seen in some of our previous problems, the subintervals do not have to be of equal width. If the widths of the subintervals are not equal, we can list them as $\Delta x_1, \Delta x_2, \Delta x_3, \ldots \Delta x_n$. In order for our limit process to work, we have to make sure that the largest width ($\max \Delta x_k$) approaches 0. In this case the definition of the integral becomes:

$$\int_a^b f(x)dx = \lim_{\max \Delta x_k \to 0} \sum_{k=1}^{n} f(x_k^*)\Delta x_k$$

Definition of the Definite Integral	If $f(x)$ is integrable on $[a, b]$, then $$\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{k=1}^{n} f(x_k)\Delta x$$ where $\Delta x = \dfrac{b-a}{n}$ and $x_k = a + \Delta x \cdot k$

You MUST know this formula. There will be questions on the AP exam that will require you to know this formula! If you prefer, you can substitute Δx and x_k into the formula and memorize this instead:

$$\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{k=1}^{n} f\left(a + \frac{b-a}{n} \cdot k\right) \cdot \frac{b-a}{n}$$

Example 1: Express $\lim_{n\to\infty}\sum_{k=1}^{n}\underbrace{(x_k^3 + x_k \sin x_k)}_{f(x_k)} \cdot \underbrace{\Delta x}_{\Delta x}$ as an integral on the interval $[0, \pi]$

a b

$$\boxed{\int_0^{\pi} (x^3 + x\sin x)\, dx}$$

$f(x) = x^3 + x\sin x$

Example 2: Express the integral as a limit of Riemann sums.

$$\int_{a \to 0}^{b \to 4} \sqrt{1+3x}\, dx$$

$$\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{k=1}^{n} f(x_k) \cdot \Delta x$$

where $\Delta x = \frac{b-a}{n}$

$x_k = a + \Delta x \cdot k$

$$\Delta x = \frac{b-a}{n} = \frac{4-0}{n} = \frac{4}{n}$$

$$x_k = a + \Delta x \cdot k = 0 + \frac{4}{n} \cdot k = \frac{4k}{n}$$

$$f(x) = \sqrt{1+3x}$$

$$f(x_k) = \sqrt{1+3\left(\frac{4k}{n}\right)} = \sqrt{1+\frac{12k}{n}}$$

$$\int_0^4 \sqrt{1+3x}\, dx = \lim_{n\to\infty} \sum_{k=1}^{n} \sqrt{1+\frac{12k}{n}} \cdot \frac{4}{n}$$

The following type of problem is usually on the multiple-choice section of the AP exam.

Example 3: Write the following limit as a definite integral. Let the lower limit of the integral be 0 ($a = 0$).

$$\lim_{n\to\infty}\sum_{k=1}^{n}\underbrace{\left(4+\frac{6k}{n}\right)^2}_{f(x_k)}\underbrace{\left(\frac{6}{n}\right)}_{\Delta x}$$

$$\Delta x = \frac{b-a}{n} = \frac{6}{n}$$

$$b-a=6$$

$$b-0=6$$

$$b=6$$

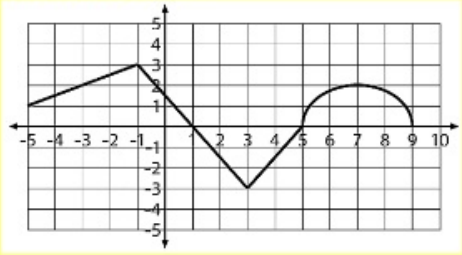
$$f(x_k) = \left(4+\underbrace{\frac{6k}{n}}_{x_k}\right)^2$$

$$x_k = a + \Delta x \cdot k = 0 + \frac{6}{n}\cdot k = \frac{6k}{n}$$

$$f(x) = (4+x)^2$$

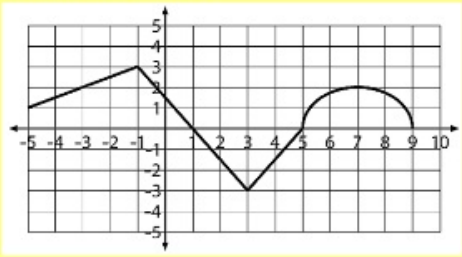
Answer: $\int_0^6 (4+x)^2\,dx$

Example 4: The graph of $f(x)$ is shown. Evaluate each integral by interpreting it in terms of area.



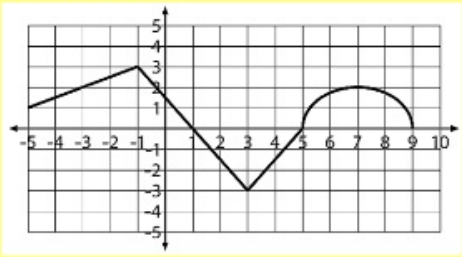
a.) $\int_{-5}^{-1} f(x)\, dx$

Example 4: The graph of $f(x)$ is shown. Evaluate each integral by interpreting it in terms of area.



b.) $\int_{-1}^{3} f(x)\ dx$

Example 4: The graph of $f(x)$ is shown. Evaluate each integral by interpreting it in terms of area.



c.) $\int_{1}^{9} f(x)\ dx$